Web-Based Salon Management and Booking System

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Approval Sheet

This design project entitled **"Web-Based Salon Management and Booking System"** Prepared by **Member A, Member B, Member C, and Member D** of the Computer Engineering Department, was examined and evaluated by the members of the Student Design Evaluation Panel and is hereby recommended for approval.

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SOFTWARE DESIGN PROJECT INFORMATION

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Project Title	Web Based Salon Management and Booking System
Project Concentration Area	Embedded Systems, Internet of Things, Machine Learning
Design Objectives	The general objective is a general statement of the objectives of the project. This may contain the features that may be included in the design. Specifically, it aims to: 1. Develop a website that: • Implements an online booking platform to eliminate overlapping bookings, long phone queues, and missed opportunities caused by manual scheduling. • Establish a structured method for feedback collection via post-visit surveys to continuously improve service. 2. Test the accuracy of the website
Constraints	
Development Cost Constraint	The total system development cost shall not exceed ₱3,000. This constraint ensures that all tools, resources, and implementation activities remain within the allocated budget.
Operational Cost Constraint	The annual hosting and domain expenses shall remain within ₱2,200 to ₱4,200. This constraint supports long-term financial sustainability of the system.

Performance Constraint	The website shall load within 3 seconds under normal operating conditions to maintain acceptable user experience and responsiveness.
	Must achieve a 75% pass rate on Google Mobile-Friendly tests (essential for spa clients on phones)
Scalability (Staff)	The system is limited to a maximum of 3 staff accounts and 1 branch.
Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.	
Manual Transaction Verification	The system does not have an automated payment API (like PayPal). It is constrained to a manual "Upload Receipt" workflow where the staff must verify GCash screenshots before confirming the appointment.
Online Communication Dependency	The system relies on internet-based communication such as Gmail for appointment confirmation and customer interaction. Staff manually communicate with customers after reviewing booking requests. Network interruptions may cause communication delays but do not affect the core functionality of the website.
Single-Branch Operation Constraint	The system is designed for a single salon branch with limited staff accounts. Multi-branch management features are not included in the current implementation. This limitation simplifies system operation without affecting website performance.
Limited Automated Scheduling Constraint	The system does not automatically assign appointment schedules. Booking requests remain in a pending status and are manually reviewed by salon staff before confirmation. This constraint supports flexible scheduling while requiring human intervention.
Standards	
Standard A – Secure Development Lifecycle (SDL)	This standard integrates security throughout the development process. Microsoft's SDL encourages threat modeling, secure coding practices, and security testing from the start. In the project, SDL practices helped identify potential threats early, ensuring the security of the final product.
Standard B – Secure	This standard implements robust authentication

Authentication and Authorization Standards	mechanisms to secure user access. Standards such as **OAuth 2.0** and **OpenID Connect** provide frameworks for secure authentication and authorization in web applications. This was crucial for the project to ensure that user data remained protected during logins.
Standard C – Data Privacy and Protection Standards	Data privacy standards ensure that user information is securely stored and protected from unauthorized use. The system applies basic data protection and confidentiality measures to safeguard customer information. This supports secure handling of system data.
Standard D – Software Quality Assurance Standards	Software Quality Assurance standards ensure that the system functions correctly through testing and validation. In this project, testing procedures were used to minimize errors and maintain reliable performance. This standard improves system accuracy and dependability.
Standard E – Usability and Accessibility Standards	Usability and accessibility standards ensure that the system is easy to understand and operate for all users. The project applies simple navigation and user-friendly interface design to improve user interaction and efficiency.

Abstract

This project addresses the need for enhanced comfort, security, and energy efficiency in residential environments by designing and implementing an intelligent home automation system leveraging Internet of Things (IoT) and machine learning techniques. The system integrates various smart sensors, actuators, and devices interconnected through a robust IoT framework. Machine learning algorithms enable the system to learn and adapt to residents' behaviors and preferences, providing personalized automation and control. Key features include automated lighting, climate control, security monitoring, and energy management. Performance evaluation through real-world scenarios demonstrates significant improvements in user convenience and energy savings. The project highlights the potential of combining IoT with machine learning to create an adaptive and efficient smart home environment, significantly enhancing the quality of life for residents by offering customized and efficient home automation solutions.

Keywords: *Home Automation, Internet of Things (IoT), Machine Learning, Smart Sensors, Energy Efficiency, Security Monitoring, Adaptive Control, Smart Home*

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CHAPTER 1: THE PROJECT AND ITS BACKGROUND

1.1 The Problem

Many small-to medium-scale salons continue to rely on manual and fragmented processes for managing daily operations such as appointment scheduling, customer records, billing, inventory monitoring, and marketing. These manual practices often result in overlapping bookings, missed or forgotten appointments, inaccurate transaction records, poor inventory tracking, and limited customer engagement. As customer demand increases and expectations shift toward digital convenience, these inefficiencies directly affect service quality, operational efficiency, and business profitability.

From an engineering design perspective, the core problem is the absence of an integrated, reliable, and scalable digital system that can automate salon operations while remaining simple enough for daily use by staff and clients. Without such a system, salon owners struggle to make data-driven decisions, customers experience inconvenience, and staff productivity is reduced. Addressing this problem requires the design and development of a web-based solution that transforms manual workflows into a centralized, automated, and user-friendly system. Therefore, there is a need for a web-based system that can centralize salon operations, reduce manual intervention, and improve service reliability and customer satisfaction.

1.2 The Client

The client for this project is **Claire Beauty Lounge**, a local salon business that provides beauty and personal care services to its customers. The salon currently operates using mostly manual procedures for booking appointments, managing customer information, tracking services rendered, and handling payments. The client seeks a digital solution to streamline internal operations, enhance customer experience, and improve accessibility through a centralized web-based platform.

Table 1-1. Client and Engineering Requirements / Considerations

Client Requirements / Considerations	Engineering Requirements /2 Considerations
The system should be able to safely store and retrieve customer information and appointment records.	The system must use a structured database to properly organize, store, and retrieve customer and appointment data, while keeping the information secure and accurate.
The system should be able to save and access customer records	A database must be used by the system in order to safely store and facilitate the

Client Requirements / Considerations	Engineering Requirements /2 Considerations
whenever needed.	retrieval of consumer information.

1.3 The Project

This project involves the design and development of a **Web Based Salon Management and Booking System** intended to automate and integrate the core operations of Claire Beauty Lounge. The system will provide features such as online appointment booking, basic functionality of customer relationship management (CRM), service and transaction tracking, inventory monitoring, and basic marketing support. By consolidating these functionalities into a single platform, the project aims to reduce manual workload, minimize operational errors, and improve overall service delivery.

1.4 Project Objectives

The general objective of this project is to design, develop, and evaluate a web-based salon management system that improves operational efficiency, customer satisfaction, and data organization for the client salon.

Specifically, the project aims to:

1. To design the overall workflow for the salon management and booking system**.**
    - Design the data model for appointments, customers, services, and transactions.
2. To develop a web-based platform that supports online booking, customer management, billing, and inventory tracking.
3. To evaluate the system's accuracy, reliability, and usability based on predefined requirements and real-world usage scenarios.

1.5 Scope and Delimitations

The scope of this project includes the design and implementation of a web-based salon management and booking system tailored for Claire Beauty Lounge. The system will cover essential salon operations such as appointment scheduling, customer record management, service tracking, billing, and basic inventory monitoring. The target users of the system include salon owners, staff, and customers accessing the booking features. The project will be implemented and

tested in a real salon setting to evaluate its effectiveness in improving workflow efficiency and customer experience.

The project is limited to the features and services identified during the requirements analysis phase. It does not account for variations in salon size, business models, or operational policies beyond the client's current setup. The system will only support the selected technology stack and will not integrate third-party platforms beyond those required for basic operation. Advanced analytics, full-scale marketing automation, and complex cybersecurity mechanisms are beyond the scope of this project. The system will provide a functional and basic user interface intended for demonstration and operational use rather than full commercial deployment.

1.6 Design Constraints

Safety (Root-Mean-Squared Error)

Functionality is an important design constraint because the system must operate accurately and safely to ensure reliable performance. The error rate defines how often the system produces incorrect outputs during operation. The error rate is obtained by computing one minus the accuracy of the system results, which measures the difference between expected and actual outputs. A higher error rate indicates lower functional performance and reduced system safety. Therefore, the design with the lowest error rate is considered the winning design because it provides the most reliable and accurate operation

Sustainability

Sustainability refers to the ability of the system to operate efficiently while minimizing energy consumption and environmental impact. The proposed system promotes sustainability by using optimized processing and reducing unnecessary resource usage during operation. Although sustainability is considered, all design alternatives consume similar resources; therefore, it did not affect the final design selection.

Public Health

Public health considers whether the system promotes user safety and prevents harmful effects during usage. The system supports public health by ensuring safe operation and minimizing user errors through accurate results and proper functionality. Since all proposed designs follow the same safety standards, this constraint did not influence the choice of the best design.

Welfare

Welfare focuses on improving user convenience, comfort, and overall benefit when interacting with the system. The system enhances user welfare by providing efficient performance and reducing manual effort. However, each design alternative provides similar user benefits; therefore, welfare was not a deciding factor.

Social

Social constraints refer to how acceptable and useful the system is within society or among intended users. The system supports social usability by being accessible and easy to understand for users with basic technical knowledge. Because all designs maintain similar usability levels, this constraint did not affect design selection.

Global

Global constraints consider compatibility with international standards and adaptability to different environments. The system follows commonly accepted technological practices, allowing possible implementation in various locations. Since all designs meet the same global considerations, this factor was not used in determining the best design.

Cultural

Cultural constraints ensure that the system respects user values, practices, and local preferences. The system design uses neutral and universally acceptable interfaces suitable for different cultural backgrounds. As cultural impact remains consistent across all designs, it did not influence the final decision.

1.7 Engineering Standards

The engineering standards provide the basis for the design, development, and evaluation of the proposed web-based salon management and booking system. Adherence to recognized software and information system standards ensures system reliability, data integrity, security, and maintainability. The project follows the applicable standards listed below.

ISO/IEC/IEEE 23026:2015 – Website Engineering and Management

ISO/IEC/IEEE 23026:2015 defines system engineering and management requirements for the life cycle of websites, including strategy, design, development, testing, and maintenance. The standard focuses on improving website usability, information accessibility, security, and maintainability.

ISO/IEC 27001 – Information Security Management

ISO/IEC 27001 provides guidelines for managing and protecting sensitive information. The system applies basic information security principles such as user authentication and controlled access to protect customer and transaction data.

ISO/IEC 12207 – Software Life Cycle Processes

This standard defines processes for software development, testing, deployment, and maintenance. The project follows ISO/IEC 12207 to ensure that the system development lifecycle is systematic and properly documented.

ISO/IEC 27002 – Information Security Controls

ISO/IEC 27002 provides best practices for implementing information security controls. The system applies selected controls such as access management and data protection to safeguard customer and transaction information.

1.8 Engineering Design Process

The development of a web-based salon management and booking system requires a methodical approach to deal with operational problems like inefficient record management and manual appointment handling. A formal framework for determining system requirements, analyzing constraints, and creating practical alternatives is provided by an engineering-based methodology, called the Engineering Design Process. This ensures that the design constraints—such as usability, accuracy, reliability, cost, data security, and scalability—are in line with the objectives of the client and engineering requirements. The procedure makes it easier to make well-informed decisions regarding everything from problem identification to system improvement, while ensuring a reliable and functional solution.

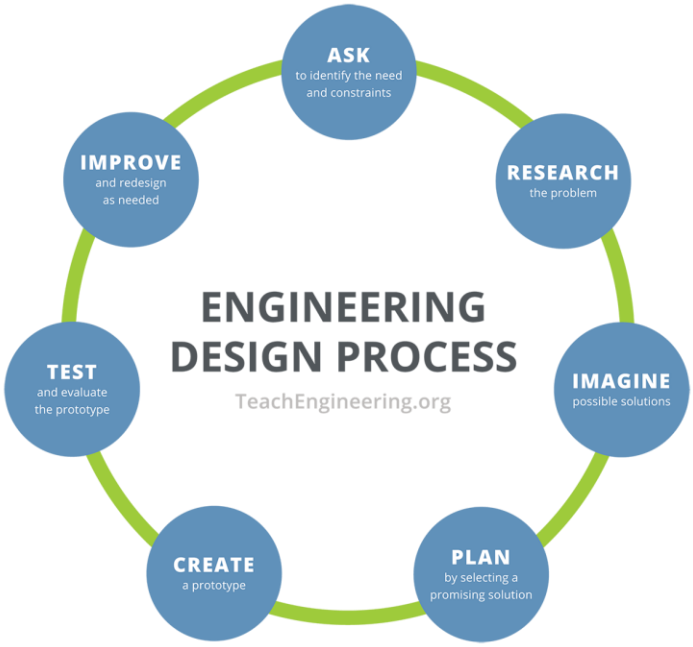

Figure 1.2 The Engineering Design Process (TeachEngineering, 2023)

Figure 1.2 illustrates the Engineering Design Process applied to the development of web-based salon management and booking systems. The process consists of a sequence of iterative stages that guide the identification of problems, development of solutions, and refinement of the final system.

1.8.1 Ask: Identifying the Need and Constraints

This stage focuses on identifying and analyzing the existing problems in salon operations. Based on gathered information, the major issues include manual appointment scheduling, inefficient customer record management, long waiting times, and overcrowding within the establishment.

Research is conducted to understand current booking workflows, customer handling processes, and limitations of existing web-based management systems. This analysis helps determine the root causes of operational inefficiencies and establishes the functional requirements needed in the proposed system.

1.8.2 Research the Problem

This stage focuses on defining the specific needs of the system and identifying its design constraints. Based on the research findings, the system must improve scheduling efficiency, organize customer records, and minimize waiting time.

Design constraints such as usability, accuracy, reliability, cost limitations, data security, and scalability are also established during this stage. These constraints ensure that the proposed system remains practical, secure, affordable, and suitable for small salon businesses.

1.8.3 Imagine: Develop Possible Solution

In this stage, possible solutions to the identified problems are generated. These include the development of a centralized web-based system capable of

handling appointment scheduling, customer records, billing, and inventory management. Multiple system features and design approaches are considered.

1.8.4 Plan: Select a Promising Solution

The plan stage involves selecting the most feasible solution based on the defined requirements and constraints. For this project, a web-based salon management and booking system is selected as the proposed solution. System architecture, database structure, and process workflows are defined during this stage to ensure consistency, reliability, and scalability.

1.8.5 Create: Build a Prototype

This stage involves the development of a functional prototype of the selected system. Core features such as online appointment booking, customer information management, transaction recording, and inventory tracking are implemented according to the system plan.

1.8.6 Test and Evaluate the Prototype

The testing stage evaluates the system's functionality, accuracy, and reliability. The system is tested against predefined requirements to verify correct scheduling behavior, data consistency, and stable performance during normal operation.

1.8.7 Improve: Redesign as Needed

The improvement stage focuses on refining the system based on testing results and evaluation outcomes. Necessary modifications are applied to address

identified limitations and enhance system performance. This iterative process continues until the system meets the defined requirements.

CHAPTER 2: SOFTWARE DESIGN

2.1 Description of the Design Solution

2.1.1 General Description

The proposed project is a practical, web-based booking and management platform designed specifically for a local salon. Instead of forcing the business to use a heavy, expensive application, this system acts as a digital storefront and an organized booking assistant that connects seamlessly with tools the staff already knows how to use, like Gmail.

The main goal of this design solution is to give the salon a professional online presence while keeping the actual day-to-day operations manual, flexible, and easy to control.

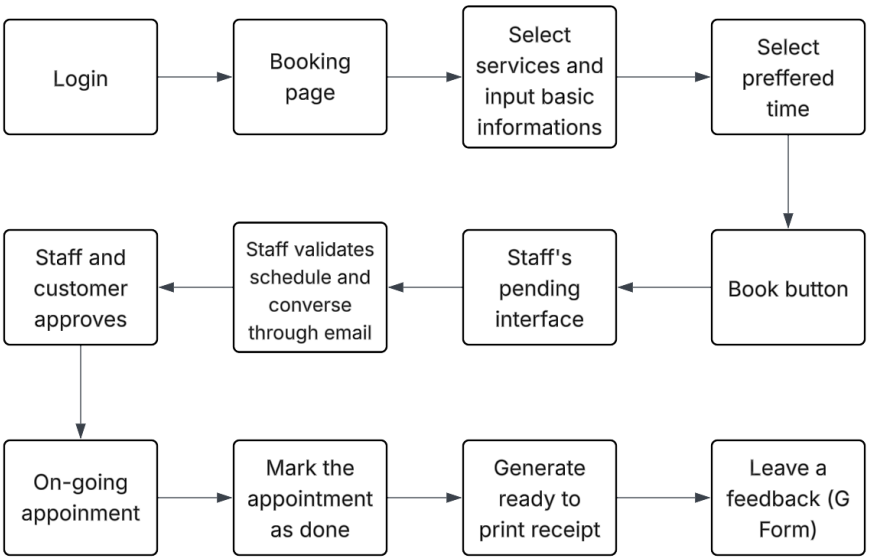

Figure 1 Illustrative Diagram of the System

The system revolves around four main features that follow the customer's journey:

- **Accessible Web Catalog:** When customers visit the website, they are greeted with a complete, structured list of the salon's services, prices, and current promotions. This replaces the need for customers to scroll through social media to figure out what the salon offers.
- **Staff "Pending" View & Email Communication:** When a customer decides to book, they fill out a form with their name, email, and desired service. This request goes straight to the staff's website interface, where it is organized and displayed as a "Pending" appointment. The staff can easily review these requests on their dashboard, check their actual schedule, and then use Gmail as their

main communication tool to chat with the customer and confirm the final time slot.

- **Printable Post-Appointment Receipt:** After the salon finishes the customer's haircut or treatment, the system provides a clean, ready-to-print digital receipt. This includes a breakdown of the services provided, the staff member's name, and the total cost, giving the salon a professional way to hand over a bill without needing expensive cashier software.
- **Structured Feedback System:** To help the salon improve its services, the website integrates Google Forms. After an appointment, customers can easily leave structured feedback (like ratings and specific comments), which is neatly organized into a spreadsheet for the salon owner to review.

2.1.2 Engineering Principles Involved

This section explains the main software engineering concepts we used to design the salon booking website. Instead of trying to build a massive, fully automated app from scratch, our project focuses on integrating existing tools (like Gmail and Google Forms) and keeping the workflow simple. By following these principles, we made sure the system is realistic to build, easy to maintain, and useful for the salon staff.

Engineering Principle

1. Modularity and Service Integration
   - **Principle:** Modularity means building a system using separate parts or external services that work together, rather than coding everything into one giant program.
   - **Explanation:** Instead of programming our own email server and review database from scratch, we applied this principle by integrating Google Workspace. We are routing the appointments to Gmail and using Google Forms for the feedback system. By letting Google handle the heavy lifting for emails and surveys, our development team can focus strictly on building a great booking interface and the printable receipt feature.
2. The KISS Principle (Keep It Simple, Stupid)
   - **Principle:** This well-known software principle states that systems work best when they are kept simple and unnecessary complexities are avoided.
   - **Explanation:** We applied this by focusing on a Minimum Viable Product (MVP). For example, our programmer advised us to use Google Forms for the feedback system first, instead of trying to code a custom widget right away. This keeps the project scope realistic, prevents over-engineering, and ensures we deliver the core features (like the service catalog and booking form) on time.

3. Human-in-the-Loop (HITL) System Design
    - **Principle:** This is a system design model where a human is required to review or make a final decision, rather than letting the software automate everything.
    - **Explanation:** Big commercial apps use complex algorithms to automatically assign booking slots. We purposely chose not to do that. Instead, we used an "Email-Based Pending Workflow." The website automatically collects and formats the booking data, but it pauses to let a human staff member review the pending request in Gmail. This gives the salon staff the final say over their schedule and allows them to talk to the client before confirming the slot.
4. Data Formatting and Usability
    - **Principle:** This principle focuses on making sure the data collected by the system is processed and presented in a way that is practical and easy for the user to read.
    - **Explanation:** We applied this directly to our printable receipt feature. The website takes the digital data from the customer's booking (such as their name, the exact services requested, and the total cost) and formats it into a clean, ready-to-print layout. This connects our digital web form to the physical, day-to-day operations of the salon, giving them a usable paper record without needing an expensive Point-of-Sale (POS) system.

2.1.3 Prior Art Analysis

This section examines existing solutions, platforms, and related methodologies currently used in the salon and beauty industry for appointment booking, customer feedback, and post-service documentation. By analyzing these existing models ranging from informal social media setups to enterprise-level booking applications and patented scheduling systems, we can identify gaps in the market and highlight the unique value proposition of the proposed design project.

- **Existing Philippine Solutions, Patent Context, and Features of Interest**

1. Philippine Patent-Based Electronic Systems (IPOPHIL Context)

Under the Intellectual Property Office of the Philippines (IPOPHL), software-related inventions such as electronic scheduling systems may be protected if they demonstrate a technical solution to a technical problem. While there are registered Philippine patents covering database management systems and automated business processes, there are currently limited publicly documented patents specifically focused on salon-oriented appointment and CRM integration systems.

Existing Philippine software-related filings typically emphasize system architecture and data processing frameworks rather than highly specialized scheduling engines. These patented systems, when present, often involve enterprise-level database structures, making them complex and costly for small, community-based salon businesses. Furthermore, most patent claims focus on transaction handling or data storage mechanisms rather than integrating structured feedback systems, printable receipts, and simplified administrative dashboards within a single lightweight platform.

In institutions such as the University of Caloocan City (UCC), particularly within the Computer Science and Information Technology programs, student capstone projects frequently involve the development of appointment systems, POS systems, and small-business management platforms.

These academic systems commonly include:

- Basic online booking forms
- Admin dashboards for monitoring requests
- Customer databases
- Transaction recording modules

While functional, these systems are often designed primarily for academic compliance and proof-of-concept purposes. Advanced features such as automated time-slot conflict detection, structured CRM analytics, integrated printable receipt generation, and scalable deployment infrastructure are usually limited or simplified. Additionally, most of these systems are not formally patented under IPOPHIL, meaning they remain academic prototypes rather than protected commercial solutions.

3. Philippine Commercial Booking Practices

Many local salons in the Philippines rely on social media platforms such as Facebook and Instagram for appointment inquiries and promotions. While these platforms are effective for visibility and client engagement, they lack a structured and automated appointment scheduling system.

Bookings are typically handled through manual back-and-forth messaging, which increases the risk of double-booking and lost inquiries. Feedback is gathered informally through comments or private messages, and there is no built-in mechanism to generate professional, formatted receipts after service completion.

4. Standalone Online Forms (Google Forms in Philippine SMEs)

Some small businesses attempt to utilize Google Forms as a free and accessible booking solution. While this approach allows for structured data collection, it does not provide a complete system infrastructure.

A standalone form:

- Does not provide a professional business website interface
- Lacks an integrated administrative “Pending View” dashboard for organized approval and scheduling
- Cannot automatically validate staff availability
- Does not generate ready-to-print, formatted receipts

As a result, while it serves as a temporary data collection tool, it does not meet the operational requirements of a fully integrated salon management system.

What sets our project apart

Our proposed project is a perfectly balanced solution tailored for small-to-medium salons. It gives the business a professional website (with a full service catalog) but keeps the actual management side simple and cost-effective by linking up with tools they already use, like Gmail and Google Forms.

Instead of forcing automated scheduling like the big apps and patented systems do, our project uses an Staff "Pending" View. This means customer booking requests go straight to a clean dashboard on the website. The staff can review the pending requests there, and then use Gmail to personally talk to the client and confirm the time slot. By adding a Ready-to-Print Post-Appointment Receipt and integrating Google Forms for feedback, we give the salon everything they need to run smoothly without the heavy costs of enterprise software.

Table 2.1 Prior Art Analysis Matrix

Design	**Features**			
	Structured Catalog & Promos	**Staff "Pending" View**	**Structured Feedback System (Forms)**	**Printable Post-Appointment Receipt**
Patented Systems	X			X

Design	Features			
	Structured Catalog & Promos	**Staff "Pending" View**	**Structured Feedback System (Forms)**	**Printable Post-Appointment Receipt**
Social Media Pages	X		X	
Google Forms			X	
Booksy	X		X	X
PROJECT	X	X	X	X

Explanation of the Matrix

Structured Catalog & Promos: Booksy, the patented systems, and our Proposed Project all give customers a clear, clickable menu of services and prices. Social Media gets a "~ (Informal)" because while it's great for posting promo photos, it doesn't have a real, structured service menu.

Staff "Pending" View: This is our project's standout feature. Instead of the system booking slots automatically, requests go to a pending dashboard where staff can review them, allowing them to manually confirm details with the client via email.

Automated Slot Assignment: We purposely left this blank for our project to give the salon staff manual control over their calendar, and to avoid the expensive development costs that come with apps like Booksy and patented enterprise systems.

Structured Feedback System: Google Forms handles this perfectly for our project, while Booksy uses its own internal system. Social Media gets a "~ (Informal)" because comments are unstructured. The cited patent is left blank because its core design focuses strictly on booking slots, not gathering customer reviews.

Printable Post-Appointment Receipt: Big apps and systems offer digital receipts, but printing them usually requires their specific, expensive hardware. Standalone forms and social media don't have receipts at all. Our project easily

turns the booking data into a clean layout that can be printed on any standard salon printer.

2.2 General System Architecture

This section explains how engineering concepts and system elements are implemented in the proposed web-based salon management and booking system. The architecture follows a **Human-in-the-Loop (HITL)** design model, where booking requests are collected digitally, but final confirmation decisions are made manually by salon staff.

The system is primarily web-based and integrates external cloud services such as Gmail and Google Forms to minimize development complexity while maintaining professional functionality.

2.2.1 Hardware Elements

- Since the system is web-based, the hardware requirements are minimal and do not require specialized computing equipment.

Hosting Server

- The web application is deployed using a shared hosting environment (Hostinger). The hosting provider supplies:
- Web server environment
- Database storage
- Domain management
- SSL certificate for HTTPS secure access
- The server stores booking requests, service data, and administrative records.

User Devices

- The system is accessible using standard internet-enabled devices:
- Smartphones
- Tablets
- Desktop or laptop computers
- Customers use these devices to browse services and submit booking requests. Salon staff use similar devices to access the administrative dashboard and review pending appointments.
- No embedded systems or specialized hardware components are required.

2.2.2 Software Elements

The system follows a simplified layered web architecture composed of frontend, backend, database, and external service integrations.

A. Frontend Layer

The frontend is built using standard web technologies such as:

- HTML
- CSS
- JavaScript (ReactJS)

The frontend provides:

- Service catalog display
- Online booking form
- Administrative “Pending Appointments” dashboard
- Printable receipt interface

The interface is responsive to ensure compatibility with both desktop and mobile browsers.

B. Backend Layer

The backend is built using standard web technologies such as:

- PHP (Laravel)

The backend manages business logic and data handling. It performs the following functions:

- Receives booking form submissions
- Stores booking data in the database
- Labels new requests as **Pending**
- Displays requests in the staff dashboard
- Updates booking status (Pending, Confirmed, Cancelled)
- Generates formatted receipt layouts

Unlike fully automated systems, the backend does **not** automatically confirm appointments or assign time slots. Final approval is handled manually by staff through email communication.

C. Database Layer

A relational database (MySQL) is used to store structured information, including:

Customers Table

- Customer ID
- Name
- Contact information

Appointments Table

- Appointment ID
- Customer ID
- Service selected
- Preferred date and time
- Status (Pending / Confirmed / Cancelled)

Services Table

- Service name
- Description
- Price

The database ensures organized record-keeping and easy retrieval for administrative review.

D. External Service Integration

To reduce complexity and development cost, the system integrates existing cloud services:

Gmail Integration

- Staff use Gmail to communicate with customers
- Booking requests are reviewed manually
- Confirmation or rescheduling is handled via email

Google Forms Integration

- Used for post-appointment feedback collection
- Responses are automatically stored in Google Sheets
- Provides structured service evaluation data

This modular integration approach supports simplicity, maintainability, and cost efficiency.

2.2.3 System Algorithm

The system follows a Human-in-the-Loop booking workflow rather than a fully automated scheduling algorithm.

Booking Process Algorithm

1. Customer selects service and submits booking form.
2. System stores booking request in database.
3. Booking status is set to **Pending**.
4. Request appears in the Staff Pending Dashboard.
5. Staff manually reviews request.
6. Staff checks actual schedule (offline or internal reference).
7. Staff confirms appointment via Gmail.
8. Staff updates booking status to **Confirmed**.

If the requested schedule is unavailable, staff communicate alternative options before confirmation.

This approach ensures manual oversight, reduces scheduling conflicts caused by outdated automated data, and preserves personalized customer interaction.

2.2.4 Data, Datasets, and Processing

A. Data Collected

The system collects the following structured data:

Customer Data

- Full name
- Email address
- Contact number

Appointment Data

- Selected service
- Preferred date and time
- Booking status

Feedback Data

- Service rating
- Customer comments

B. Data Processing Scheme

The data processing flow includes three stages:

Pre-processing

- Validate required form fields
- Check correct input formats (email, date, time)

Processing

- Store request as Pending
- Display in administrative dashboard
- Format booking details for receipt generation

Post-processing

- Update booking status after staff action
- Store confirmed appointments for records
- Collect feedback through Google Forms

C. Receipt Data Formatting

- After service completion, the system retrieves booking data and formats it into a structured printable receipt containing:
- Customer name
- Service availed
- Staff name
- Service cost
- Total amount
- This bridges digital booking records with the salon's day-to-day operational needs.

2.3 Design Alternatives

2.3.1 Rationale for Design Alternatives

The design alternatives were developed to explore different technology implementations for a web-based salon booking and management system while considering cost, development complexity, implementation speed, scalability, and usability for small-to-medium salon operations.

The objective of comparing these alternatives is not only to produce a working system but also to determine the most practical technology stack for the target client. Small salon businesses typically operate with limited technical resources and budget constraints. Therefore, selecting an appropriate development technology is important to ensure that the system remains affordable, maintainable, and easy to deploy.

By evaluating multiple programming approaches, the project minimizes the risk of selecting an overly complex solution while ensuring that the chosen design provides an effective balance between performance, development effort, and usability. The alternatives mainly differ in the programming language used for the backend system and the level of development complexity required to implement the web application.

2.3.1 Design Alternative A: Python-Based Web System

A. Engineering Principles of Alternative

Server-Side Web Application Development

This approach utilizes Python as the primary backend programming language for handling server-side logic. Python frameworks such as Flask or Django manage form submissions, database communication, and server responses.

Framework-Based System Architecture

Python web frameworks provide structured architecture that organizes routing, business logic, and database operations within a modular environment.

Database-Driven Application Logic

The system relies on a relational database to store booking requests, customer data, and appointment records, which are processed by the backend application.

B. Architecture of Design Alternative

In this design:

Customer submits booking request through the website interface
Python backend processes the booking request
Booking information is stored in the database
System sends confirmation or notification email
Staff can review booking records through the administrative interface

Key Components

Frontend (HTML/CSS/JavaScript)
→ Booking Form
→ Service Catalog Interface

Backend (Python – Flask/Django)
→ Request handling
→ Business logic processing
→ Email notification service

Database (MySQL)
→ Customers
→ Appointments
→ Services

C. Constraints

Constraint A – Additional Framework Setup
Python web frameworks require server configuration and environment setup.

Constraint B – Moderate Development Complexity
Backend logic must be developed and maintained by programmers familiar with Python frameworks.

Constraint C – Hosting and Deployment Requirements
The system requires a web server capable of running Python-based applications.

iv. Evaluation Results

Although this design is fast and scalable, it does not match the salon's preference for manual review and email-based communication. Because of this, it is less suitable for the target users.

2.3.2 Design Alternative B: HTML, CSS, and JavaScript Web System

i. Engineering Principles of Alternative

Native Web Development

This design uses standard web technologies including HTML for page structure, CSS for visual styling, and JavaScript for client-side interactivity. These technologies form the foundation of modern web development.

Lightweight Web Application Design

The system focuses on creating a responsive and user-friendly website interface while minimizing unnecessary backend complexity.

User-Centered Interface Design

The interface is designed to be simple and accessible so that salon staff and customers can easily interact with the system without requiring advanced technical knowledge.

ii. Architecture of Design Alternative

Workflow:

Customer accesses the salon website
Customer selects services and fills out the booking form
JavaScript validates the form input
Booking request is sent to the backend server
Staff reviews the request through the admin interface
Confirmation email is sent to the customer
Feedback is collected through online forms

Key Components

Frontend
→ HTML
→ CSS
→ JavaScript

Backend **Support**
→ PHP or Node.js for simple server-side processing

Database
→ MySQL

iii. Constraints

Constraints

Constraint A – Limited Advanced Automation

The system relies on basic web technologies and may require additional backend tools if advanced automation features are needed.

Constraint B – Dependency on Web Hosting Services

The system requires a hosting provider to store and serve the website.

Constraint C – Browser Compatibility Considerations

The interface must be tested across multiple browsers to ensure consistent functionality.

iv. Evaluation Results

This design provides the best balance between implementation speed, cost efficiency, technical complexity, and usability. Because HTML, CSS, and JavaScript are standard technologies for web development, the system can be developed faster and maintained more easily compared to other alternatives. Therefore, this approach is considered the most suitable implementation for the project.

2.3.3 Design Alternative C: C++ Based Web System

i. Engineering Principles of Alternative

High-Performance Backend Programming

This design explores the use of C++ as the primary backend programming language for handling web requests and system processing.

Compiled System Architecture

C++ applications are compiled into executable programs, allowing efficient resource usage and high processing performance.

Custom Web Framework Integration

Specialized C++ web frameworks such as Crow or CppCMS are used to build web server functionality and manage HTTP requests.

ii. Architecture of Design Alternative

In this design:

Customer submits booking request through the website
C++ backend processes incoming requests
Booking data is stored in the database
Server handles confirmation notifications
Staff reviews appointment requests through the interface

Key Components

Frontend (HTML/CSS/JavaScript)
→ Booking Form
→ Service Catalog

Backend (C++ Web Framework)
→ Request processing
→ System logic

Database (MySQL)
→ Customer records
→ Appointment data

iii. Constraints

Constraints

Constraint A – High Technical Complexity

C++ web development requires specialized programming knowledge and advanced system configuration.

Constraint B – Longer Development Time

Developing web applications using C++ frameworks generally requires more development effort compared to standard web technologies.

Constraint C – Higher Maintenance Difficulty

Future modifications and updates may require developers experienced in C++ programming.

2.4 Standards Involved in the Design

This section presents the standards followed by the design, including their references.
Matrix may be used to show how standards are used in each specific design.

Table xx Summary of Standards Involved in the Alternatives

Standard	Brief Description	DESIGNS		
		DESIGN A	DESIGN B	DESIGN C

IEC 60950	Product Safety Standard for electronic and computing products.	Used in enclosure, power supply leakage, ESD, wiring, and connectors.		
Philippine National Standards for Drinking Water (PNSDW)	Standards for drinking water quality, water sampling and examination and evaluation.	Used in conditional statements to determine if water is drinkable.		
IEEE 1309-2013	Standard for Calibration of Electromagnetic Field Sensors and Probes	NA	NA	Calibration of sensors used in detecting heavy metals.
IEEE 1858-2016	IEEE Standard for Camera Phone Image Quality	Reference for image processing cameras.	NA	NA

CHAPTER 3: DESIGN TRADEOFFS

3.1 Summary of Constraints

Explain table xx below in this paragraph.

Table xx Summary of Design Constraints

Designs	Constraints				
	COST	Implementation speed	Technical Complexity	Maintenance Effort	Staff Ease of Use
Design A – Python web implementation	2	2	2	1	3
Design B – HTML/CSS/Java Script Website	1	1	1	1	2
Design C – C++ Web System	2	2	3	2	1

Synthesize for the next section.

3.2 Trade-offs

Table xx Preference and Importance of Constraints

Constraints	Preference	Importance (raw)	% Importance

Explain the use of Pareto Multi-Criteria Decision Making (MCDM).

$$Minimization = 9 \times \left(\frac{Max\ Value - Raw\ Value}{Max\ Value - Min\ Value}\right) + 1$$ Equation No. xx

$$Maximization = 9 \times \left(\frac{Raw\ Value - Min\ Value}{Max\ Value - Min\ Value}\right) + 1$$ Equation No. xx

3.2.1 Tradeoff 1: Constraint A (Metric)

3.2.1.1 Design 1: Normalization of Constraint A (Metric)
<Introduce>

Table xx Evaluation of Three Design Alternatives based on Constraint A

Design	Constraint (Metric)

<Analyze>

3.2.1.2 Design 2: Normalization of Constraint A (Metric)

Table xx Evaluation of Three Design Alternatives based on Constraint B

Design	Constraint (Metric)

3.2.1.3 Design 3: Normalization of Constraint A (Metric)

Table xx Evaluation of Three Design Alternatives based on Constraint C

Design	Constraint (Metric)

3.2.2 Tradeoff 2: Constraint B (Metric)

3.2.2.1 Design 1: Normalization of Constraint B (Metric)
3.2.2.2 Design 2: Normalization of Constraint B (Metric)
3.2.2.3 Design 3: Normalization of Constraint B (Metric)

3.2.3 Tradeoff 3: Constraint C (Metric)

3.2.3.1 Design 1: Normalization of Constraint C (Metric)
3.2.3.2 Design 2: Normalization of Constraint C (Metric)
3.2.3.3 Design 3: Normalization of Constraint C (Metric)

3.2.4 Tradeoff 4: Constraint D (Metric)

3.2.4.1 Design 1: Normalization of Constraint D (Metric)
3.2.4.2 Design 2: Normalization of Constraint D (Metric)
3.2.4.3 Design 3: Normalization of Constraint D (Metric)

3.2.5 Tradeoff 5: Constraint E (Metric)

3.2.5.1 Design 1: Normalization of Constraint E (Metric)
3.2.5.2 Design 2: Normalization of Constraint E (Metric)
3.2.5.3 Design 3: Normalization of Constraint E (Metric)

3.3 Summary of the Normalized Values of the Three Designs

Designs	Constraints				
	Constraint A (metric)	Constraint B (metric)	Constraint C (metric)	Constraint D (metric)	Constraint E (metric)
Design A					
Design B					
Design C					

3.4 Designers Raw Ranking for the Three Designs

Table xx Designers Raw Ranking for the Three Designs

Decision Criteria	Criterion's Importance		Ability to Satisfy Criterion		
	Scale (0-10)	Percentage (%)	Design A	Design B	Design C

3.5 Sensitivity Analysis
3.6 Influence of the Design Tradeoffs in the Final Design

CHAPTER 4: FINAL DESIGN

4.1 Final Design
4.1.1 Software Application
4.1.2 Hardware Design
4.2 Test Procedures and Evaluation
4.2.1 Test Procedures
4.2.2 Test Evaluation
4.3 Test and Evaluation Results
4.3.1 Test Results
4.3.2 Evaluation Results
4.4 Conclusion
4.5 Impact of the Design
4.5.1 Societal
Target UN SDG.
4.5.2 Ethical
In compliance with known ethical codes/standards
4.5.3 Legal
National / Intl Laws
4.6 Sustainability Plan

CHAPTER 5: BUSINESS PLAN AND MODEL

5.1 Business Plan
5.1.1 Executive Summary
5.1.2 General Company Description
5.1.3 Products and Services Offered
5.1.4 Marketing Plan
5.1.5 Marketing Strategy
5.2 Business Model
5.3 Intellectual Property (IP) Reports

REFERENCES

Note: This must be done using APA format. Check the guide for more details: https://www.scribbr.com/apa-style/apa-seventh-edition-changes/

Covey, S. R. (2013). *The 7 habits of highly effective people: Powerful lessons in personal change*. Simon & Schuster.

APPENDICES

Include standards preview, certification from experts/clients, code snippets, patent reports, and other long and detailed documents here. Format is as follows below:

APPENDIX A: TITLE OF THE SECTION

<figure>

Note: No figure number. Standards must be followed with a paragraph explaining its contents and purpose.